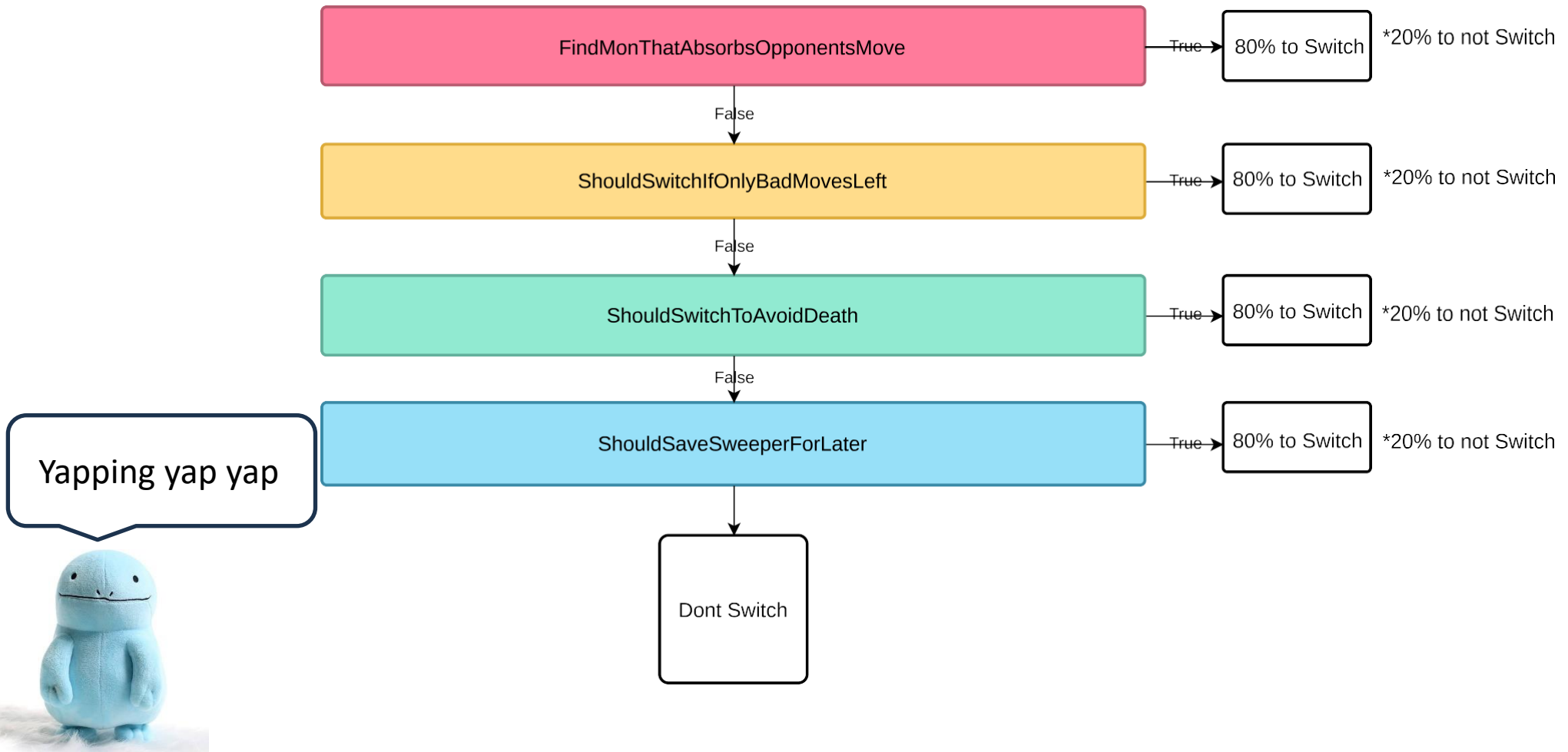


RR Mid Turn Switch Mechanics

General:

There are multiple reasons for why the AI would consider a hard switch, those are expressed by boolean functions (true or false as a result). The result of those depends on the current situation on the field and the mons the AI has in the party. Once one of these is true the AI will perform a switch and ignore the functions coming after. If all those functions are false, the AI won't switch. The order in which the AI checks those is the following (there are more functions, however I will just go over the most common ones):



Miscellaneous

Damage: The RR AI doesn't roll for damage (I am not sure about the rounding) and calculates damage for mons currently out and mons in the party differently:
 $\text{Dmg_field} = \text{max roll} * 0.93$ (damage you do to the AI on the field and vice versa)
 $\text{Dmg_party} = \text{max roll} * 0.96$ (damage your mon does against a mon in the party)
A move OHKO/2HKO/3HKO/4HKO's according to that (based on current HP)

SwitchInScores: Govern the Post-KO switch-ins, but also play a big role for mid turn switches. I won't go over those in detail, the calculator gives them fairly accurate. For edge cases (f.e. 3HKO vs 4HKO) you might wanna calculate the damage explicitly, ideally you want to put yourself into positions that eliminate disambiguity when it comes to those.

Switch in Flags (only those that are not self-explanatory)

Outspeeds : AI mon is faster (duh), speed tie means AI is also faster

Resist all Moves: The AI mon is not 3HKO'd by you

Walls foe: Compares your relevant (!) offensive stat(s) to the AI's corresponding defensive stat(s), it means the AI's defense(s) is greater than your offense(s)

AI Kill Flag: Due to a bug(?) the flag for AI OHKO'ing the player does not exist, so in fact AI switches (post and mid turn) are mostly just motivated by the defensive capabilities and speed.

CoolDown: Means that if the AI just switched in (either after a KO or a hard switch that is not type absorption related) it can't switch out the following turn.
Note: If AI hard switched but the player switched/pivoted into a different mon, this does not apply.

AntiAbuse and Switch: Usually if you switched out a mon and switch back into it the following turn, the AI reads your input and predicts, however the decision to switch out goes before the AntiAbuse check, so don't worry about it when forcing switches.

Pivot: If the AI decides to switch out a faster mon, it will use its pivot move to do so. In general pivoting has a different logic than switching though.

PredictedMove: For certain functions the move the AI predicts you to go for matters, a killing move usually is the same as the predicted move, however status moves, stat lowering moves and knock off can have higher scores than the strongest (non killing) move and would therefore be the AI predicted move (avoid having those on the moveset if you want to make sure nothing unexpected happens)

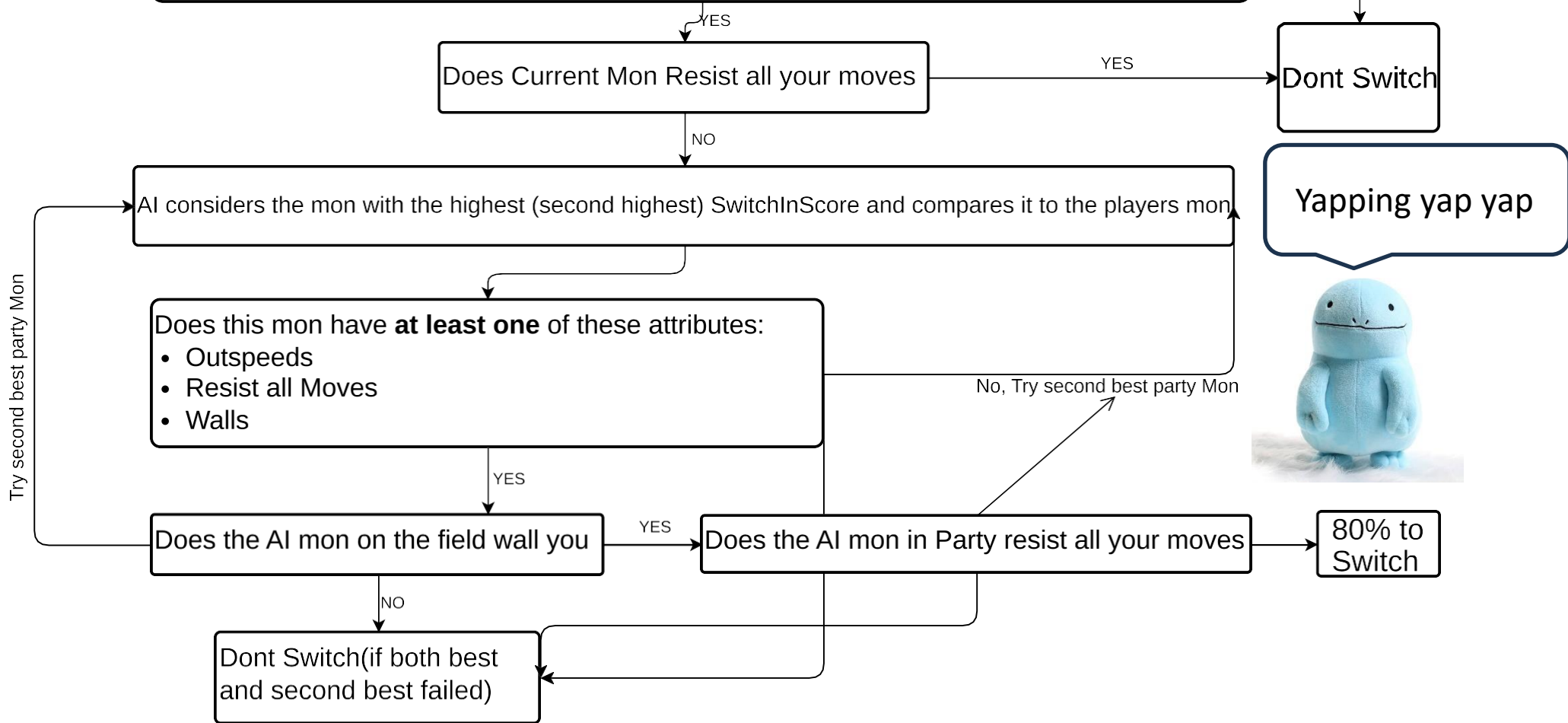
Dont Switch: Doesn't necessarily mean don't switch at all, rather it means consider the next function, only if SaveSweeperForLater results in don't switch, it means AI certainly doesn't switch.



SwitchIfOnlyBadMovesLeft

All of these need to be **true** in order for the AI to **consider** a switch:

- AI mon not on cooldown
- AI mon doesnt have a viable status move (including hazards, setup moves)
- AI cant knock off an item
- AI mon cant 3HKO your current mon out (calcs with leftovers if applicable)



Note: If the conditions at the top are met for an AI mon in a double battle (in relation to both your mons on the field), AI switching out becomes very likely (details unclear)

Note: This is usually also the reason an encored mon will switch out

FindMonThatAbsorbsOpponentsMove

General:

Type absorption is the first reason why the AI would consider a switch.

They don't follow the typical cooldown, meaning they can trigger even if the AI just switched in another mon.

On top of that, the type absorbing mon itself can immediately switch out again.

Only back and forth switches between multiple type absorbing mons are prohibited (this even applies in RR)

The AI almost always switches out into a type absorbing mon, if the move it predicts you to go for (= strongest move if you are an all out attacker and none of your moves have a sideeffect) against the current mon is of the corresponding type, cases where it won't switch:

AI has a kill on you

AI has a slow kill on you (and you don't KO first)

AI already invested into setup

AI switched that mon for type absorption before --> 25% chance to not switch

The last move you went for wasn't the move the AI predicted you to do

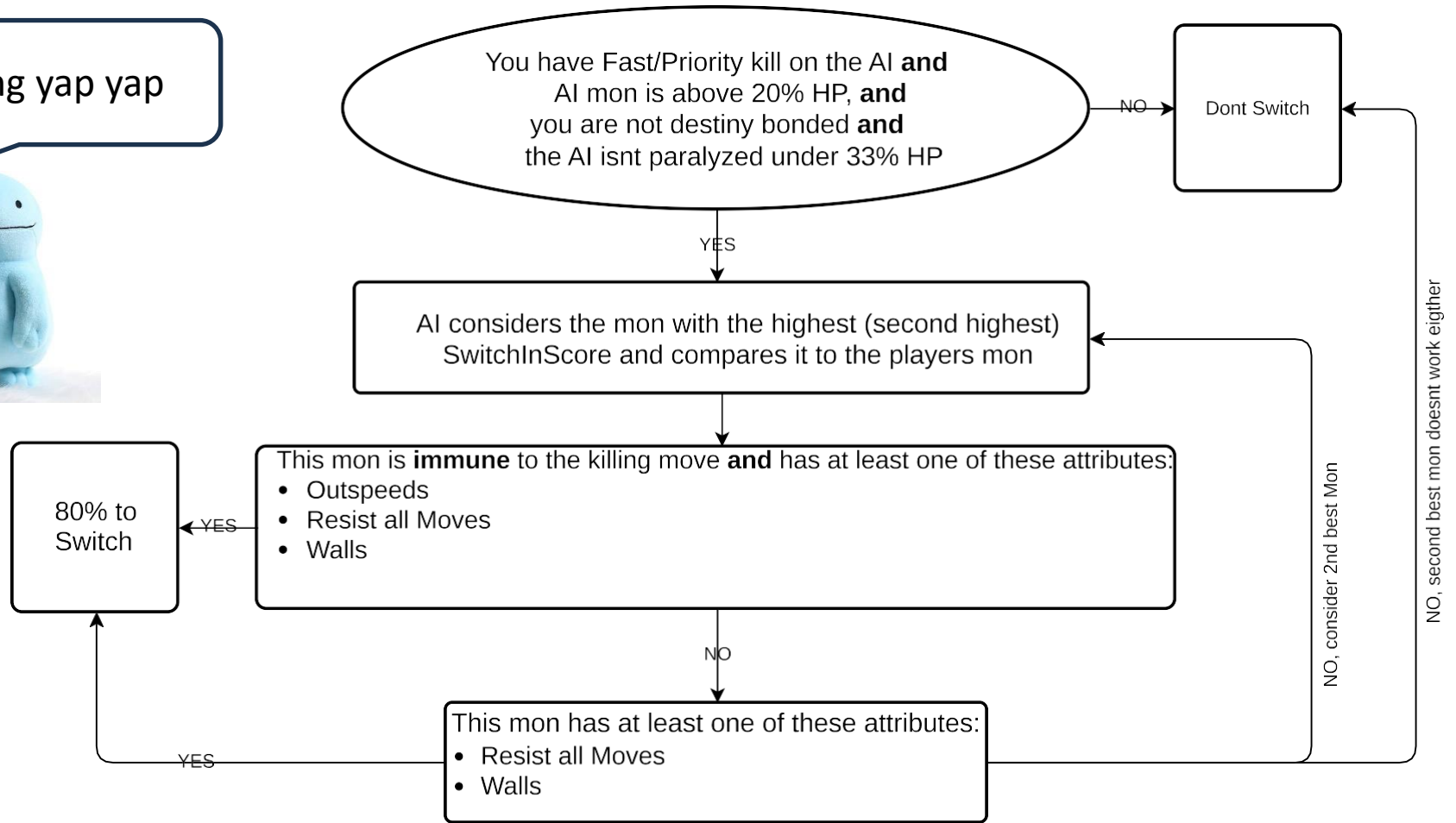
Your mon is not of the damager class --> 50% chance to not switch

Yapping yap yap



ShouldSwitchToAvoidDeath

Yapping yap yap



Cases where you have a slow kill (independent of HP) or a fast kill on the AI (below 20%) are covered by SaveSweeperForLater

Yapping yap yap

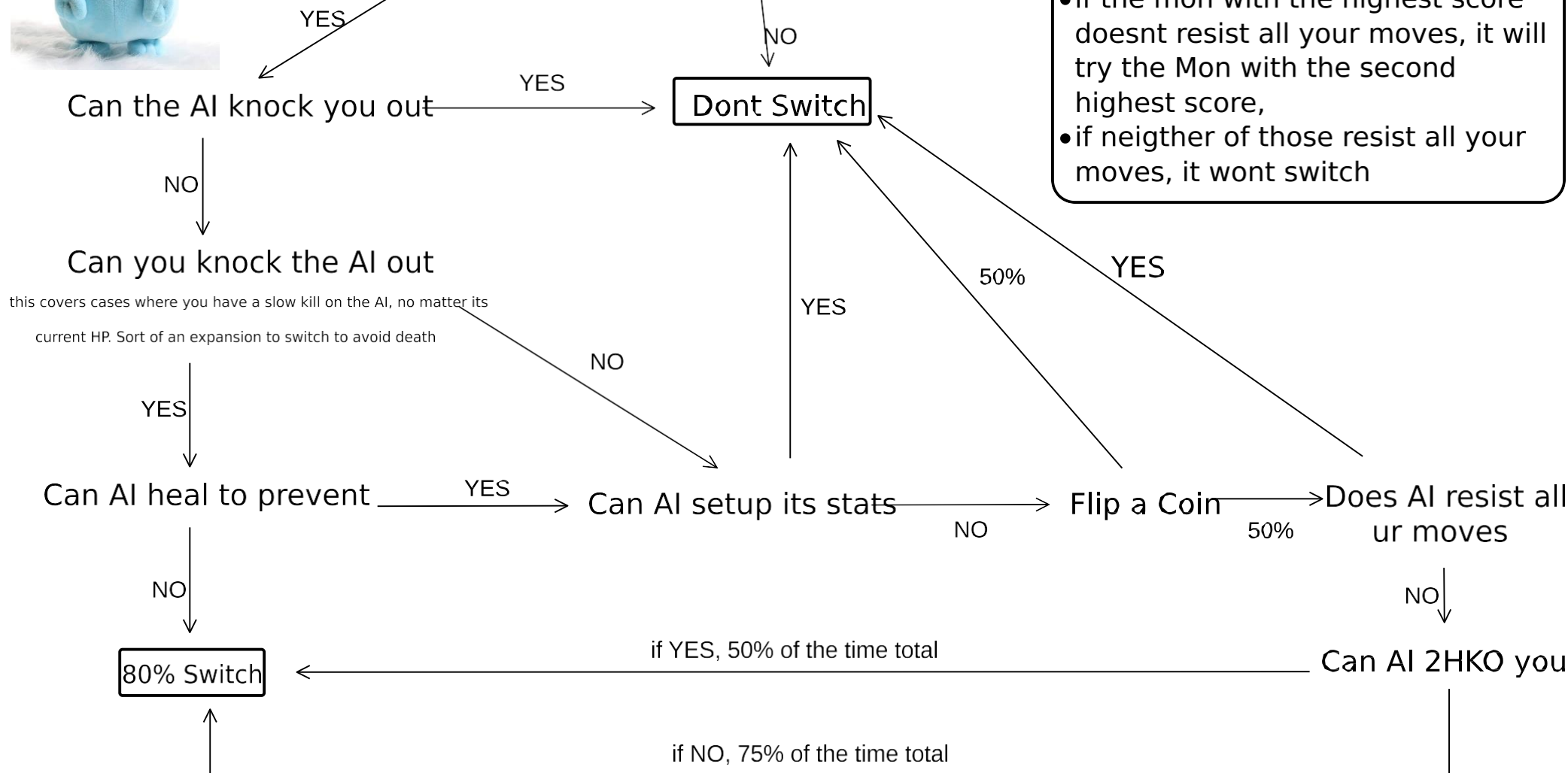
ShouldSaveSavesweeperforlater



Single Battle and
AI mon isnt on switch cooldown and
AI mon is of the damager Class (mostly means its not a hazard setter)*

Switch:

- If Switch: AI will go into the mon with highest score that resists all your moves (3hko+)
- if the mon with the highest score doesnt resist all your moves, it will try the Mon with the second highest score,
- if neighter of those resist all your moves, it wont switch



*a hazard setter that already used its hazard move is not considered a hazard setter anymore.

If the mon has the ability toxic debris or hazard setting